

Assignment 8: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

NOTE: For this assignment, you can skip tests for any of the assumptions required to run the models.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.2      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Spring2026_Shiyi
```

```
library(lubridate)
```

```
library(agricolae)
```

```
here()
```

```
## [1] "/home/guest/EDE_Spring2026_Shiyi"
```

```
NTL.chem.phys <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"), stringsAsFactors = TRUE)
```

```
NTL.chem.phys$sampldate <- mdy(NTL.chem.phys$sampldate)
```

```
#2
```

```
mytheme <- theme_classic(base_size = 14) +  
  theme(axis.text = element_text(color = "black"), legend.position = "top")
```

```
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: mean lake temperature recorded during July DO NOT change with depth across all lakes Ha: mean lake temperature recorded during July change with depth across all lakes
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
```

```
July.temp <- NTL.chem.phys %>%  
  filter(month(sampldate) == 7) %>%  
  select(lakename, year4, daynum, depth, temperature_C) %>%  
  drop_na()
```

```
#5
```

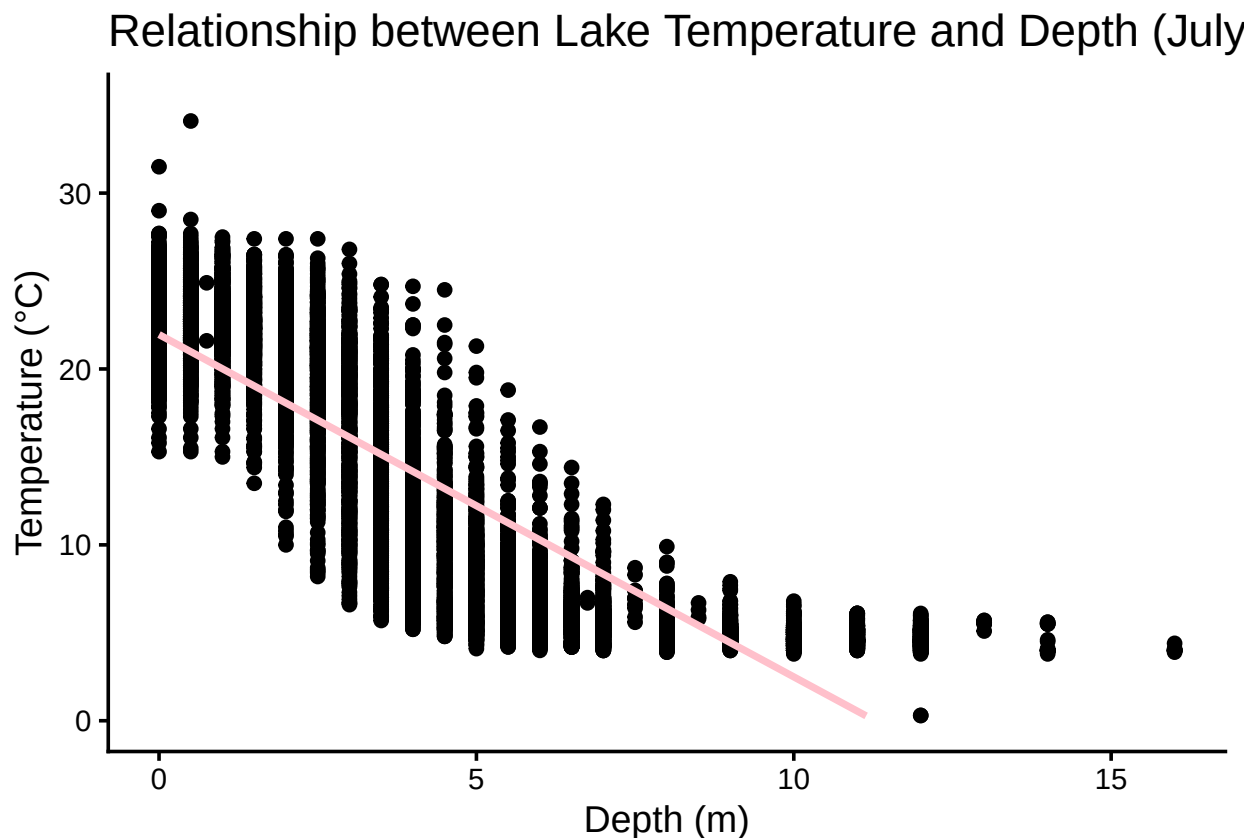
```
temp.depth.plot <- ggplot(July.temp, aes(x = depth, y = temperature_C)) +  
  geom_point() +  
  geom_smooth(method = "lm", color = "pink", se = FALSE) +
```

```
ylim(0,35)+
labs(x = "Depth (m)",
     y = "Temperature (°C)",
     title = "Relationship between Lake Temperature and Depth (July)")

print(temp.depth.plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The scatterplot shows a negative relationship between depth and temperature. As depth increases, lake temperature decreases, indicating that deeper water is colder. The distribution of points suggests that this relationship is approximately linear, as the points follow a general downward trend and align reasonably well with the fitted linear regression line. However, there is some variability in temperature at shallower depths, indicating that the relationship is not perfectly linear.

7. Perform a linear regression to test the relationship and display the results.

```
#7
temp.depth.lm <- lm(data=July.temp, temperature_C ~ depth)
summary(temp.depth.lm)

##
## Call:
## lm(formula = temperature_C ~ depth, data = July.temp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: A linear regression was conducted to examine the relationship between lake temperature and depth in July. The results showed a significant negative relationship between depth and temperature (p value: $< 2.2e-16$), indicating that temperature decreases as depth increases. The model explained approximately 73.87% of the variability in temperature (Multiple R-squared: 0.7387), based on 9726 degrees of freedom. The regression coefficient for depth was -1.94621, meaning that for every 1 meter increase in depth, temperature is predicted to decrease by approximately 1.94621°C.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
temp.full <- lm(data = July.temp, temperature_C ~ year4 + daynum + depth)
temp.AIC <- stepAIC(temp.full) #Based on AIC model selection, the full model including year4, daynum, and
```

```
## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4    1         101 141788 26070
## - daynum   1         1237 142924 26148
## - depth    1      404475 546161 39189
```

```
print(temp.AIC)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = July.temp)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##   -8.57556      0.01134      0.03978     -1.94644
```

```
#10
temp.multi <- lm(data = July.temp, temperature_C ~ year4 + daynum + depth)
summary(temp.multi)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = July.temp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: ased on the AIC model selection, the final set of explanatory variables includes year4, daynum, and depth, as removing any of these variables increased the AIC value. This multiple regression model explains approximately 74.12% of the observed variance in temperature (Multiple R-squared: 0.7412). Compared to the simple regression model using only depth (Multiple

R-squared: 0.7387), this represents a slight improvement in explanatory power ($0.7412 - 0.7387 = 0.0025$). This suggests that including temporal variables such as year and day of year provides additional, though relatively small, explanatory value beyond depth alone.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
temp.anova.aov <- aov(data = July.temp, temperature_C ~ lakename)
summary(temp.anova.aov)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
temp.anova.lm <- lm(data = July.temp, temperature_C ~ lakename)
summary(temp.anova.lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = July.temp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake        -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake   -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: A one-way ANOVA was conducted to test whether mean lake temperature differed among lakes in July. The results showed a significant effect of lake on temperature (F Value = 50, $p < 2e-16$), indicating that mean temperatures differ significantly among lakes. The linear model results further support this finding, showing that several lakes have significantly different mean temperatures compared to the reference lake. This may also suggest that lake-specific characteristics influence temperature patterns during July.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

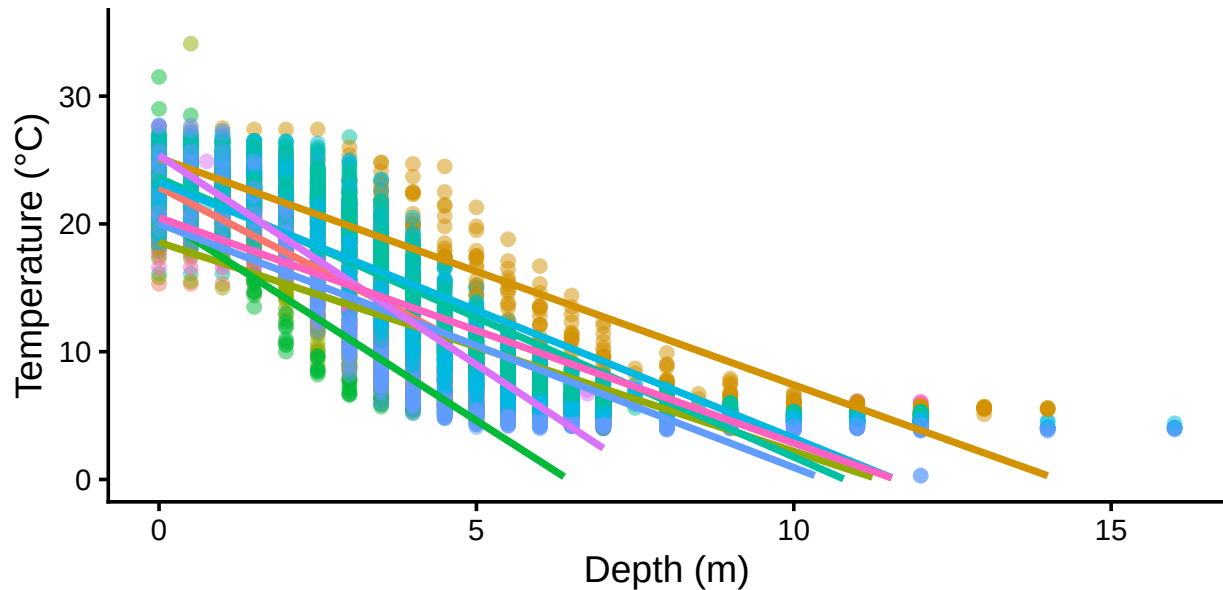
```
#14.
temp.depth.lake.plot <- ggplot(July.temp, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  ylim(0, 35) +
  labs(title = " Relationships Between Temperature and Depth in July by Lakes",
        x = "Depth (m)",
        y = "Temperature (°C)",
        color = "Lake")
print(temp.depth.lake.plot)

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Relationships Between Temperature and Depth in July b

Central Long Lake East Long Lake Paul Lake Tuesday Lake W
Crampton Lake Hummingbird Lake Peter Lake Ward Lake



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(temp.anova.aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = July.temp)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000

## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Based on Tukey's HSD test, Paul Lake and Ward Lake have mean temperatures that are not significantly different from Peter Lake ($p > 0.05$). No lake has a mean temperature that is statistically distinct from all other lakes, as each lake has at least one pairwise comparison that is not statistically significant.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we were only comparing Peter Lake and Paul Lake, a two-sample t-test could be used to determine whether their mean temperatures differ significantly. This turkey test is appropriate because it compares the means of a continuous variable between two independent groups

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
CramptonLake.WardLake.temp <- July.temp %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

t.test(data=CramptonLake.WardLake.temp, temperature_C ~ lakename)

##
## Welch Two Sample t-test
##
```

```

## data:  temperature_C by lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
##  -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862

```

Answer: A two-sample t-test was conducted to compare mean July temperatures between Crampton Lake and Ward Lake. The results showed no significant difference between the two lakes ($t = 1.18$, $p = 0.2649$), indicating that their mean temperatures are statistically similar. The 95% confidence interval $(-0.68, 2.47)$ includes zero, further supporting that there is no significant difference in mean temperature between the two lakes. This result is consistent with the findings from the Tukey's HSD test in part 16, which also indicated no significant difference between Crampton Lake and Ward Lake.